Inequalities — problem 7.50

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Let $0 = a_0 < a_1 < \ldots < a_n$ and $a_{i+1} - a_i \leq 1$ for $0 \leq i \leq n-1$. Then,

$$\left(\sum_{i=0}^{n} a_i\right)^2 \geq \sum_{i=0}^{n} a_i^3.$$